

PHYS 486 - HW 06

1) $\Psi(x,t) = \sum_n c_n \Psi_n(x) e^{-iE_n t/\hbar}$ with $E_n = \left(n + \frac{1}{2}\right) \hbar \omega$

$$\langle X \rangle = \langle \Psi(x,t) | \hat{x} | \Psi(x,t) \rangle$$

Noting that $\hat{x} = \sqrt{\frac{\hbar}{2m\omega}} (a + a^\dagger)$ and $\Psi_n(x) = \frac{(a^\dagger)^n}{\sqrt{n!}} |0\rangle$

$$\begin{aligned} \langle X \rangle &= \sum_n c_n^* e^{+iE_n t/\hbar} \langle \Psi_n | \sqrt{\frac{\hbar}{2m\omega}} (a + a^\dagger) | \Psi \rangle \sum_m c_m e^{-iE_m t/\hbar} \\ &= \sum_{n,m} c_n^* c_m e^{-i(E_m - E_n)t/\hbar} \sqrt{\frac{\hbar}{2m\omega}} \left(\langle \Psi_n | a | \Psi_m \rangle + \langle \Psi_n | a^\dagger | \Psi_m \rangle \right) \\ &= \sum_{n,m} c_n^* c_m e^{-i(E_m - E_n)t/\hbar} \sqrt{\frac{\hbar}{2m\omega}} \left(\sqrt{m} \langle \Psi_n | \Psi_{m-1} \rangle + \sqrt{m+1} \langle \Psi_n | \Psi_{m+1} \rangle \right) \\ &= \sum_{n,m} c_n^* c_m e^{-i(m-n)\omega t} \sqrt{\frac{\hbar}{2m\omega}} \left(\sqrt{m} \delta_{m,n+1} + \sqrt{m+1} \delta_{m,n-1} \right) \\ &= \sum_{n=0}^{\infty} c_n^* c_{n+1} e^{-i\omega t} \sqrt{\frac{\hbar}{2m\omega}} \sqrt{n+1} + \underbrace{\sum_{n=1}^{\infty} c_n^* c_{n-1} e^{i\omega t} \sqrt{\frac{\hbar}{2m\omega}} \sqrt{n}} \end{aligned}$$

Note that summation starts from $n=1$ since in $\delta_{m,n-1} = 0$ for $n=0$.

$\Rightarrow$ Changing the index from $n \rightarrow n+1$ on second summation.

$$\begin{aligned} &= \sum_{n=0}^{\infty} \sqrt{\frac{\hbar}{2m\omega}} \sqrt{n+1} \left(c_n^* c_{n+1} e^{-i\omega t} + c_{n+1}^* c_n e^{i\omega t} \right) \\ &= \sum_{n=0}^{\infty} \sqrt{\frac{\hbar}{2m\omega}} \sqrt{n+1} e^{i\phi} c_n^* c_{n+1} e^{-i(\omega t + \phi)} + \sqrt{\frac{\hbar}{2m\omega}} \sqrt{n+1} e^{-i\phi} c_{n+1}^* c_n e^{i(\omega t + \phi)} \end{aligned}$$

$$= \frac{C}{2} e^{i(\omega t + \phi)} + \frac{C^*}{2} e^{-i(\omega t + \phi)} = \operatorname{Re} \left\{ C e^{i(\omega t + \phi)} \right\}$$

$$= C \cos(\omega t + \phi),$$

$$\text{where } \frac{C}{2} = \sum_n \sqrt{\frac{\hbar}{2m\omega}} e^{-i\phi} c_{n+1}^* c_n \sqrt{n+1}$$

$$C e^{i\phi} = \sum_n \sqrt{\frac{2\hbar}{m\omega}} \sqrt{n+1} c_{n+1}^* c_n$$

2.a)

$$\bullet \quad \underset{\substack{\text{operator } x}}{P_x} (\psi(x)) = P(x \psi(x)) = -x \psi(-x)$$

$$x P(\psi(x)) = x (\psi(-x)) = x \psi(-x)$$

$$P_x (\psi(x)) = -x P(\psi(x)) = -x \psi(-x) \Rightarrow \boxed{P_x = -x P}$$

$$\bullet \quad P_p (\psi(x)) = P \left(-i\hbar \frac{\partial \psi(x)}{\partial x} \right) = -i\hbar \frac{\partial \psi(-x)}{\partial (-x)} = i\hbar \frac{\partial \psi(-x)}{\partial x}$$

$$p P(\psi(x)) = -i\hbar \frac{\partial}{\partial x} (\psi(-x)) = -i\hbar \frac{\partial \psi(-x)}{\partial x}$$

$$P_p (\psi(x)) = -p P(\psi(x)) = i\hbar \frac{\partial \psi(-x)}{\partial x} \Rightarrow \boxed{P_p = -p P}$$

$$\bullet \quad a^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \left(x - i \frac{p}{m\omega} \right)$$

$$P a^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \left(P_x - i \frac{P_p}{m\omega} \right) = \sqrt{\frac{m\omega}{2\hbar}} \left(-P_x + i \frac{P_p}{m\omega} \right) \Rightarrow \text{using previous results}$$

$$= -P \sqrt{\frac{m\omega}{2\hbar}} \left(x - i \frac{p}{m\omega} \right) = -P a^\dagger \Rightarrow \boxed{a^\dagger P = -P a^\dagger}$$

2.b)

$$|n\rangle = \frac{(a^\dagger)^n}{\sqrt{n!}} |0\rangle$$

Note from the previous part that

$$P (a^\dagger)^n = P a^\dagger (a^\dagger)^{n-1} = -a^\dagger P a^\dagger (a^\dagger)^{n-2} = (-1)^2 (a^\dagger)^2 P (a^\dagger)^{n-2} = \dots = (-1)^n (a^\dagger)^n P$$

$$P|n\rangle = P \frac{(a^\dagger)^n}{\sqrt{n!}} |0\rangle = (-1)^n \frac{(a^\dagger)^n}{\sqrt{n!}} P|0\rangle$$

Since $|0\rangle \propto e^{-m\omega x^2/2\hbar}$, $P|0\rangle = (+1)|0\rangle = |0\rangle$: parity operator leaves ground state unchanged

$$P|n\rangle = (-1)^n \frac{(a^\dagger)^n}{\sqrt{n!}} |0\rangle = (-1)^n |n\rangle$$

→ Multiply $P|n\rangle = (-1)^n |n\rangle$ with $\langle x|$ on both sides

$$\langle x|P|n\rangle = (-1)^n \langle x|n\rangle = (-1)^n \psi_n(x)$$

$$\text{Using } \langle x|P = \langle -x|$$

$$\langle x|P|n\rangle = \langle -x|n\rangle = \boxed{\psi_n(-x) = (-1)^n \psi_n(x)}$$

⇒ Energy eigenfunction for even n have even parity ⇒ $\psi(-x) = \psi(x)$

" " for odd n have odd parity ⇒ $\psi(-x) = -\psi(x)$

3)

$$H = \begin{pmatrix} a & 0 & b \\ 0 & c & 0 \\ b & 0 & a \end{pmatrix}$$

3.a) For H to be Hermitian, we need $(H^*)^T = H^\dagger = H$

$$(H^*)^T = \begin{pmatrix} a^* & 0 & b^* \\ 0 & c^* & 0 \\ b^* & 0 & a^* \end{pmatrix} \Rightarrow$$

$$\left. \begin{array}{l} a^* = a \\ b^* = b \\ c^* = c \end{array} \right\} \begin{array}{l} a, b, c \text{ must be real for} \\ H \text{ to be Hermitian.} \end{array}$$

3.b)

$$|S(0)\rangle = \begin{pmatrix} 0 \\ d \\ 0 \end{pmatrix}$$

For $|S(0)\rangle$ to be normalized $\Rightarrow \langle S(0)|S(0)\rangle = |d|^2 = 1 \Rightarrow |d| = 1$

One can choose $d = e^{i\phi}$, $\phi \in \mathbb{R}$

$\rightarrow$ To time evolve the state, we apply the unitary operator $U(t) = e^{-iHt/\hbar}$:

$$|S(t)\rangle = U(t)|S(0)\rangle = e^{-iHt/\hbar} \begin{pmatrix} 0 \\ d \\ 0 \end{pmatrix} = \sum_{n=0}^{\infty} \left(\frac{-it}{\hbar} \right)^n \frac{H^n}{n!} \begin{pmatrix} 0 \\ d \\ 0 \end{pmatrix}$$

Check $H|S(0)\rangle = \begin{pmatrix} a & 0 & b \\ 0 & c & 0 \\ b & 0 & a \end{pmatrix} \begin{pmatrix} 0 \\ d \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ cd \\ 0 \end{pmatrix}$

$$H^2|S(0)\rangle = \begin{pmatrix} a & 0 & b \\ 0 & c & 0 \\ b & 0 & a \end{pmatrix} \begin{pmatrix} 0 \\ cd \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ c^2d \\ 0 \end{pmatrix}$$

$$H^n|S(0)\rangle = c^n \begin{pmatrix} 0 \\ d \\ 0 \end{pmatrix}, //$$

$$|S(t)\rangle = \sum_{n=0}^{\infty} \frac{(-itc/\hbar)^n}{n!} \begin{pmatrix} 0 \\ d \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ e^{-itc/\hbar} d \\ 0 \end{pmatrix}$$

→ Alternatively noticing that $\begin{pmatrix} 0 \\ d \\ 0 \end{pmatrix}$ is an eigvec of H with eigval c ,
one can directly assert $e^{-iHt/\hbar} |x\rangle = e^{-i\lambda t/\hbar} |x\rangle$

$$\hookrightarrow |S(t)\rangle = e^{-ict/\hbar} |S(0)\rangle$$

3.c) Notice $S(0) = \begin{pmatrix} d \\ 0 \\ 0 \end{pmatrix}$ is not an eigenstate of H . So, we shall try to write it in terms of eigstates of H .

$$\det \{H - \lambda I\} = \begin{vmatrix} a-\lambda & 0 & b \\ 0 & c-\lambda & 0 \\ b & 0 & a-\lambda \end{vmatrix} = (a-\lambda)(c-\lambda)(a-\lambda) - b(c-\lambda)b = (c-\lambda)(a^2 - b^2 - 2a\lambda + \lambda^2)$$

$$= (c-\lambda)(\lambda - (a+b))(\lambda - (a-b))$$

• $\lambda_1 = c$ with eigvec. $|v_1\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$

• $\lambda_2 = a+b \Rightarrow \begin{pmatrix} a & 0 & b \\ 0 & c & 0 \\ b & 0 & a \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} (a+b)x \\ (a+b)y \\ (a+b)z \end{pmatrix} \Rightarrow \begin{cases} ax + bz = ax + bx \\ cy = ay + by \\ bx + az = az + bz \end{cases} \begin{cases} x = z \\ \text{Let } y=0 \end{cases}$
(although no requirement on y)

$\lambda_2 = a+b$ with eigvec $|v_2\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \Leftarrow$ Note that we've normalized x_2

• $\lambda_3 = a-b = \begin{pmatrix} a & 0 & b \\ 0 & c & 0 \\ b & 0 & a \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} (a-b)x \\ (a-b)y \\ (a-b)z \end{pmatrix} \Rightarrow \begin{cases} ax + bz = ax - bx \\ cy = ay - by \\ bx + az = az - bz \end{cases} \begin{cases} x = -z \\ \text{Let } y=0 \end{cases}$

$\lambda_3 = a-b$ with eigvec. $|v_3\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$

$$\text{Now } |S(0)\rangle = \begin{pmatrix} d \\ 0 \\ 0 \end{pmatrix} = \sqrt{2} d (|v_2\rangle + |v_3\rangle)$$

$$|S(t)\rangle = e^{-iHt/\hbar} |S(0)\rangle = \sqrt{2} d \left[e^{-iHt/\hbar} |v_2\rangle + e^{-iHt/\hbar} |v_3\rangle \right]$$

$$= \sqrt{2} d \left(e^{-i\lambda_2 t/\hbar} |v_2\rangle + e^{-i\lambda_3 t/\hbar} |v_3\rangle \right)$$

$$= d e^{-iat/\hbar} \left[e^{-ibt/\hbar} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + e^{ibt/\hbar} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \right]$$

$$= d e^{-iat/\hbar} \begin{pmatrix} e^{-ibt/\hbar} + e^{ibt/\hbar} & 0 \\ e^{-ibt/\hbar} - e^{ibt/\hbar} & 0 \end{pmatrix} = d e^{-iat/\hbar} \begin{pmatrix} \cos(bt/\hbar) \\ 0 \\ -i\sin(bt/\hbar) \end{pmatrix} //$$

4) To solve S.E. inside the box where $V(\vec{r}) = 0$, we use separation of variables and assume wavefunction can be written as the product of three one-dim. functions:

$$\Psi(x, y, z) = X(x) Y(y) Z(z)$$

Inside the box $-\frac{\hbar^2}{2m} \nabla^2 \Psi(x, y, z) = E \Psi(x, y, z)$

$$-\frac{\hbar^2}{2m} \left[\frac{1}{X(x)} \frac{\partial^2 X(x)}{\partial x^2} + \frac{1}{Y(y)} \frac{\partial^2 Y(y)}{\partial y^2} + \frac{1}{Z(z)} \frac{\partial^2 Z(z)}{\partial z^2} \right] = E = E_x + E_y + E_z$$

↳ Since eq. should be satisfied for any (x, y, z) ; the problem separates into 3 indep. 1D infinite square wells:

$$-\frac{\hbar^2}{2m} \frac{\partial^2 X(x)}{\partial x^2} = E_x X(x) \Rightarrow X(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n_x \pi x}{a}\right), \text{ similarly for } y, z$$

with $E_x = \frac{n_x^2 \pi^2 \hbar^2}{2ma^2}, \quad n_x = 1, 2, \dots$

→ Total wavefunction is then

$$\Psi_{n_x, n_y, n_z}(x, y, z) = \left(\frac{2}{a}\right)^{3/2} \sin\left(\frac{n_x \pi x}{a}\right) \sin\left(\frac{n_y \pi y}{a}\right) \sin\left(\frac{n_z \pi z}{a}\right)$$

with $E_{n_x, n_y, n_z} = \frac{\pi^2 \hbar^2}{2ma^2} (n_x^2 + n_y^2 + n_z^2)$

where n_x, n_y, n_z are 3 indep. quantum numbers.

4.b) $E_{n_x, n_y, n_z} = E_0 (n_x^2 + n_y^2 + n_z^2)$ with $E_0 = \frac{\pi^2 \hbar^2}{2ma^2}$

Level	States (n_x, n_y, n_z)	$n_x^2 + n_y^2 + n_z^2$	Energy	Degeneracy
1	(1, 1, 1)	1 + 1 + 1	3 E_0	1
2	(2, 1, 1), (1, 2, 1), (1, 1, 2)	1 + 1 + 4	6 E_0	3
3	(2, 2, 1), (2, 1, 2), (1, 2, 2)	1 + 4 + 4	9 E_0	3
4	(3, 1, 1), (1, 1, 3), (1, 3, 1)	1 + 1 + 9	11 E_0	3
5	(2, 2, 2)	4 + 4 + 4	12 E_0	1
6	(1, 2, 3), (1, 3, 2), (2, 1, 3) (2, 3, 1), (3, 1, 2), (3, 2, 1)	1 + 4 + 9	14 E_0	6

4.c)

Level 7 : (3, 2, 2) $\rightarrow$ 17 E_0

Level 8 : (4, 1, 1) $\rightarrow$ 18 E_0

Level 9 : (3, 3, 1) $\rightarrow$ 19 E_0

Level 10 : (4, 2, 1) $\rightarrow$ 21 E_0

Level 11 : (3, 3, 2) $\rightarrow$ 22 E_0

Level 12 : (4, 2, 2) $\rightarrow$ 24 E_0

Level 13 : (4, 3, 1) $\rightarrow$ 26 E_0

For Level 14: $E = (3^2 + 3^2 + 3^2)E_0 = (5^2 + 1^2 + 1^2)E_0 = 27 E_0$

There are 2 entirely different sets of quantum numbers:

$$(3, 3, 3) \text{ and } (5, 1, 1), (1, 5, 1), (1, 1, 5)$$

Total degeneracy for E_{14} is $1 + 3 = 4$

$\rightarrow$ What makes this case interesting is that in all of the previous levels degeneracy arose from spatial symmetry (the fact that you can permute x, y, z axes); however for E_{14} , the degeneracy arises from mathematical coincidence where two completely unrelated sets of quantum numbers happened to yield the same energy.